

Supplementary materials

Table I presents the predicted values, FEA results, and prediction errors for each model and objective in Section III-B (Error Analysis and Propagation Effects) of Section III (Optimization Results and Error Analysis).

TABLE I
PREDICTED VALUES AND ERRORS OF EACH OBJECTIVE FUNCTION BY DIFFERENT MODELS AT DIFFERENT STAGES

Method	Value	Stage 1				Stage 2				Stage 3			
		T_{ripple}	T_{avg} (N·m)	P_{loss} (W)	f	T_{ripple}	T_{avg} (N·m)	P_{loss} (W)	f	T_{ripple}	T_{avg} (N·m)	P_{loss} (W)	f
BP	Pred.	0.079	40.228	118.559	0.143	0.367	40.732	107.677	0.430	0.198	38.727	110.211	0.263
	FEA	0.368	40.610	127.964	0.490	0.368	40.656	112.957	0.447	0.469	38.970	119.245	0.589
	Error	-78.56%	-0.94%	-7.35%	-70.75%	-0.39%	0.19%	-4.67%	-3.79%	-57.71%	-0.62%	-7.58%	-55.28%
	Pred.	-0.064	42.994	150.403	0.054	0.162	45.450	141.895	0.266	0.063	48.084	159.066	0.185
	FEA	0.163	45.240	163.085	0.328	0.163	45.390	147.086	0.281	0.122	47.922	165.642	0.270
	Error	-139.39%	-4.96%	-7.78%	-83.66%	-0.26%	0.13%	-3.53%	-5.48%	-48.10%	0.34%	-3.97%	-31.47%
	Pred.	0.148	42.886	119.926	0.205	3.3×10^{-5}	0.260	0.651	1×10^{-6}	1.1×10^{-4}	1.5×10^{-4}	0.050	2.6×10^{-4}
	FEA	0.180	43.600	135.791	0.281	0.179	43.617	132.879	0.272	0.269	43.538	130.186	0.365
	Error	-17.91%	-1.64%	-11.68%	-26.86%	-100.02%	-99.40%	-99.51%	-100.00%	-99.96%	-100.00%	-99.96%	-99.93%
CNN	Pred.	-0.019	38.558	124.400	0.062	0.412	35.665	110.062	0.522	0.395	34.486	93.205	0.464
	FEA	0.445	40.615	134.182	0.592	0.444	40.670	118.319	0.547	0.285	37.204	100.461	0.344
	Error	-104.19%	-5.06%	-7.29%	-89.49%	-7.24%	-12.31%	-6.98%	-4.49%	38.28%	-7.31%	-7.22%	34.99%
	Pred.	0.150	31.691	104.456	0.242	0.281	33.253	109.958	0.393	0.281	29.395	98.341	0.387
	FEA	0.358	32.698	111.821	0.487	0.358	32.674	115.395	0.497	0.499	31.332	100.570	0.622
	Error	-58.14%	-3.08%	-6.59%	-50.44%	-21.40%	1.77%	-4.71%	-21.03%	-43.64%	-6.18%	-2.22%	-37.88%
	Pred.	0.133	29.262	94.350	0.211	0.224	30.649	87.226	0.283	0.320	32.010	85.331	0.375
	FEA	0.356	33.017	97.930	0.444	0.356	33.036	92.643	0.429	0.193	32.406	90.408	0.245
	Error	-62.66%	-11.37%	-3.66%	-52.46%	-37.06%	-7.23%	-5.85%	-33.99%	65.73%	-1.22%	-5.62%	52.95%
Kriging	Pred.	0.008	34.888	100.421	0.050	0.202	38.081	111.211	0.275	0.245	39.189	96.659	0.275
	FEA	0.203	37.959	128.879	0.326	0.202	38.058	116.036	0.288	0.245	39.257	101.341	0.287
	Error	-96.02%	-8.09%	-22.08%	-84.63%	-0.10%	0.06%	-4.16%	-4.82%	0.14%	-0.17%	-4.62%	-4.27%
	Pred.	0.172	28.382	57.528	0.158	0.375	32.459	85.064	0.433	0.126	34.017	84.258	0.143
	FEA	0.376	32.374	95.280	0.463	0.377	32.382	88.299	0.445	0.184	33.920	87.512	0.217
	Error	-54.18%	-12.33%	-39.62%	-65.91%	-0.40%	0.24%	-3.66%	-2.53%	-31.30%	0.29%	-3.72%	-34.13%
	Pred.	0.004	36.412	83.231	-0.013	0.210	37.857	91.439	0.229	0.080	40.667	93.771	0.072
	FEA	0.212	37.729	106.739	0.275	0.210	37.852	95.862	0.242	0.091	40.650	97.446	0.095
	Error	-98.03%	-3.49%	-22.02%	-104.66%	-0.30%	0.01%	-4.61%	-5.41%	-12.59%	0.04%	-3.77%	-24.47%
LSTM	Pred.	0.111	41.080	120.754	0.179	0.215	38.605	101.856	0.259	0.095	41.367	102.847	0.109
	FEA	0.217	38.454	115.320	0.300	0.216	38.545	103.662	0.266	0.101	41.465	107.284	0.128
	Error	-48.67%	6.83%	4.71%	-40.16%	-0.75%	0.16%	-1.74%	-2.73%	-6.04%	-0.24%	-4.14%	-14.51%
	Pred.	0.144	43.709	122.012	0.202	0.170	43.186	107.221	0.192	0.099	45.888	114.690	0.116
	FEA	0.173	43.003	121.366	0.236	0.171	43.117	108.888	0.199	0.128	46.881	123.435	0.166
	Error	-16.33%	1.64%	0.53%	-14.58%	-1.00%	0.16%	-1.53%	-3.54%	-22.23%	-2.12%	-7.08%	-29.79%
	Pred.	0.175	36.633	93.504	0.205	0.366	32.672	86.464	0.426	0.139	35.212	86.263	0.154
	FEA	0.368	32.562	95.772	0.454	0.367	32.603	88.796	0.434	0.179	35.430	89.667	0.207
	Error	-52.25%	12.50%	-2.37%	-54.74%	-0.29%	0.21%	-2.63%	-1.88%	-22.61%	-0.61%	-3.80%	-25.68%
RSM	Pred.	0.040	35.678	95.834	0.067	0.190	38.504	105.436	0.243	0.069	41.379	110.632	0.102
	FEA	0.193	38.383	122.652	0.294	0.191	38.492	110.326	0.257	0.103	41.295	114.589	0.151
	Error	-79.45%	-7.05%	-21.87%	-77.22%	-0.54%	0.03%	-4.43%	-5.80%	-33.20%	0.20%	-3.45%	-32.84%
	Pred.	0.050	48.003	139.998	0.118	0.126	48.784	123.804	0.151	0.086	51.151	133.593	0.118
	FEA	0.128	48.497	141.443	0.205	0.126	48.764	129.488	0.167	0.068	51.143	139.619	0.114
	Error	-60.77%	-1.02%	-1.02%	-42.60%	-0.24%	0.04%	-4.39%	-9.79%	26.98%	0.02%	-4.32%	2.98%
	Pred.	-0.009	45.686	118.649	0.008	0.150	47.155	109.550	0.149	0.057	47.977	124.699	0.082
	FEA	0.151	46.952	126.594	0.199	0.149	47.178	115.095	0.164	0.136	47.975	129.060	0.183
	Error	-107.31%	3.58%	-5.65%	-96.14%	0.46%	-0.05%	-4.82%	-8.91%	-58.40%	0.00%	-3.38%	-55.10%
SVM	Pred.	0.119	44.175	120.697	0.166	0.124	44.255	108.865	0.138	0.099	45.814	113.376	0.112
	FEA	0.125	44.105	125.754	0.188	0.126	44.218	112.560	0.151	0.093	45.737	117.711	0.119
	Error	-5.14%	0.16%	-4.02%	-11.59%	-1.54%	0.08%	-3.28%	-8.44%	5.69%	0.17%	-3.68%	-5.69%
	Pred.	0.148	41.748	118.517	0.210	0.154	41.580	108.576	0.190	0.126	43.234	113.084	0.159
	FEA	0.156	41.470	125.364	0.240	0.157	41.539	112.343	0.204	0.125	43.149	117.419	0.172
	Error	-4.98%	0.67%	-5.46%	-12.42%	-1.52%	0.10%	-3.35%	-6.62%	0.13%	0.20%	-3.69%	-7.30%
	Pred.	0.117	43.127	123.101	0.179	0.191	42.822	106.253	0.216	0.114	43.043	112.716	0.146
	FEA	0.195	42.669	122.247	0.266	0.194	42.761	109.778	0.229	0.168	42.736	117.138	0.221
	Error	-39.72%	1.07%	0.70%	-32.73%	-1.57%	0.14%	-3.21%	-5.97%	-32.08%	0.72%	-3.78%	-33.71%
No Surrogate	FEA	0.097	46.165	130.643	0.156	0.097	46.284	120.390	0.127	0.087	46.542	119.305	0.111
	FEA	0.081	46.615	140.145	0.162	0.084	46.746	129.144	0.134	0.084	46.746	129.144	0.134
	FEA	0.113	39.309	111.383	0.167	0.113	39.390	102.392	0.142	0.103	41.270	101.531	0.116

The predicted values, corresponding FEA results, and calculated prediction errors for all models under each sampling condition are presented in Table II of Section IV (Influence of Sampling Number on Surrogate Model Fitting Accuracy).

TABLE II
PREDICTED VALUES AND ERRORS OF EACH OBJECTIVE FUNCTION FOR DIFFERENT SURROGATE MODELS WITH VARYING SAMPLING NUMBERS

Method	Value	200 Samples				400 Samples				800 Samples			
		T_{ripple}	T_{avg} (N·m)	P_{loss} (W)	f	T_{ripple}	T_{avg} (N·m)	P_{loss} (W)	f	T_{ripple}	T_{avg} (N·m)	P_{loss} (W)	f
BP	Pred.	0.017	45.955	128.643	0.063	0.067	35.804	92.031	0.086	0.124	44.517	111.456	0.144
	FEA	0.158	46.382	135.551	0.236	0.348	32.701	95.959	0.432	0.180	43.172	115.319	0.226
	Error	-89.37%	-0.92%	-5.10%	-73.45%	-80.83%	9.49%	-4.09%	-80.14%	-31.06%	3.12%	-3.35%	-36.37%
	Pred.	-0.238	40.687	118.354	-0.214	0.034	49.691	161.709	0.149	-0.260	41.759	122.031	-0.235
	FEA	0.289	46.803	149.160	0.418	0.093	49.995	164.036	0.219	0.211	37.727	120.739	0.314
	Error	-182.33%	-13.07%	-20.65%	-151.20%	-63.30%	-0.61%	-1.42%	-32.02%	-223.22%	10.69%	1.07%	-175.05%
	Pred.	0.079	44.940	156.095	0.215	0.026	47.759	150.473	0.122	-0.022	36.394	95.460	-0.008
	FEA	0.191	45.413	164.593	0.361	0.154	46.719	154.520	0.283	0.344	33.827	97.597	0.424
	Error	-58.81%	-1.04%	-5.16%	-40.32%	-83.03%	2.23%	-2.62%	-56.98%	-106.46%	7.59%	-2.19%	-101.86%
CNN	Pred.	-0.169	31.442	104.265	-0.113	0.025	38.425	122.893	0.108	-0.119	27.380	86.912	-0.078
	FEA	0.286	36.306	102.068	0.354	0.124	43.299	119.634	0.176	0.375	32.437	95.400	0.463
	Error	-159.30%	-13.40%	2.15%	-131.95%	-79.73%	-11.26%	2.72%	-38.61%	-131.76%	-15.59%	-8.90%	-116.86%
	Pred.	-0.054	35.712	111.995	0.007	0.136	32.848	105.887	0.223	-0.086	28.649	91.083	-0.038
	FEA	0.169	39.794	113.146	0.231	0.370	32.731	114.097	0.507	0.342	33.747	97.086	0.421
	Error	-132.30%	-10.26%	-1.02%	-96.92%	-63.13%	0.36%	-7.20%	-56.08%	-125.28%	-15.11%	-6.18%	-109.14%
	Pred.	0.151	31.500	109.564	0.258	0.127	29.170	93.983	0.204	0.090	30.188	96.533	0.163
	FEA	0.359	32.331	115.154	0.500	0.335	32.603	98.622	0.425	0.372	32.495	96.276	0.461
	Error	-57.96%	-2.57%	-4.85%	-48.36%	-62.03%	-10.53%	-4.70%	-51.97%	-75.78%	-7.10%	0.27%	-64.63%
Kriging	Pred.	-0.017	36.214	83.148	-0.035	0.011	35.238	91.485	0.026	0.074	48.653	129.980	0.112
	FEA	0.211	37.801	106.607	0.274	0.212	37.888	121.875	0.316	0.084	48.645	135.647	0.139
	Error	-107.85%	-4.20%	-22.01%	-112.71%	-94.73%	-6.99%	-24.94%	-91.74%	-11.91%	0.02%	-4.18%	-19.49%
	Pred.	0.017	35.022	76.948	0.006	0.004	35.549	76.542	0.026	0.114	45.289	136.990	0.200
	FEA	0.192	38.310	109.614	0.258	0.211	37.801	106.607	0.274	0.108	45.184	142.048	0.207
	Error	-91.13%	-8.58%	-29.80%	-102.52%	-98.29%	-5.96%	-28.20%	-109.56%	6.10%	0.23%	-3.56%	-3.64%
	Pred.	-0.035	36.227	80.991	-0.061	0.104	35.323	87.238	0.117	0.074	48.653	129.980	0.112
	FEA	0.211	37.801	106.607	0.274	0.269	36.571	105.725	0.345	0.084	48.645	135.647	0.139
	Error	-116.54%	-4.16%	-24.03%	-122.41%	-61.27%	-3.41%	-17.49%	-65.91%	-11.91%	0.02%	-4.18%	-19.49%
LSTM	Pred.	0.117	45.178	127.380	0.176	0.144	45.345	125.261	0.199	0.116	43.217	120.409	0.169
	FEA	0.180	44.424	119.650	0.230	0.172	44.955	126.308	0.237	0.186	42.374	120.313	0.252
	Error	-35.07%	1.70%	6.46%	-23.43%	-16.74%	0.87%	-0.83%	-15.97%	-37.66%	1.99%	0.08%	-33.08%
	Pred.	0.134	40.305	106.987	0.172	0.079	46.508	127.973	0.127	0.107	43.915	124.716	0.167
	FEA	0.212	39.947	110.091	0.270	0.154	46.664	124.634	0.199	0.203	42.844	123.605	0.278
	Error	-36.76%	0.90%	-2.82%	-36.34%	-48.54%	-0.33%	2.68%	-36.52%	-47.17%	2.50%	0.90%	-39.99%
	Pred.	0.117	45.178	127.380	0.176	0.123	42.989	128.022	0.199	0.108	44.327	123.611	0.162
	FEA	0.180	44.424	119.650	0.230	0.189	42.063	122.275	0.264	0.199	43.408	121.426	0.263
	Error	-35.07%	1.70%	6.46%	-23.43%	-35.22%	2.20%	4.70%	-24.47%	-45.54%	2.12%	1.80%	-38.45%
RSM	Pred.	0.072	44.873	127.733	0.129	0.032	36.206	74.482	-0.005	0.010	49.654	138.748	0.058
	FEA	0.179	43.722	121.761	0.240	0.211	37.801	106.607	0.274	0.133	48.033	131.824	0.187
	Error	-59.80%	2.63%	4.90%	-46.15%	-84.94%	-4.22%	-30.13%	-101.83%	-92.16%	3.38%	5.25%	-68.69%
	Pred.	-0.110	32.082	54.671	-0.190	0.032	34.806	70.904	-0.077	0.010	49.654	138.748	0.058
	FEA	0.211	37.801	106.607	0.274	0.211	37.801	106.607	0.274	0.133	48.033	131.824	0.187
	Error	-151.98%	-15.13%	-48.72%	-169.36%	-115.29%	-7.92%	-33.49%	-128.08%	-92.16%	3.38%	5.25%	-68.69%
	Pred.	0.072	44.873	127.733	0.129	0.019	48.223	130.167	0.054	0.010	49.654	138.748	0.058
	FEA	0.179	43.722	121.761	0.240	0.139	48.051	129.261	0.187	0.133	48.033	131.824	0.187
	Error	-59.80%	2.63%	4.90%	-46.15%	-86.36%	0.36%	0.70%	-71.14%	-92.16%	3.38%	5.25%	-68.69%
SVM	Pred.	0.111	47.670	137.487	0.181	0.115	34.464	90.996	0.146	0.146	40.295	108.045	0.188
	FEA	0.114	46.743	135.326	0.185	0.245	34.647	99.363	0.313	0.146	40.270	113.639	0.204
	Error	-2.52%	1.98%	1.60%	-1.92%	-52.95%	-0.53%	-8.42%	-53.29%	-0.11%	0.06%	-4.92%	-7.83%
	Pred.	0.133	49.995	169.388	0.279	0.088	47.403	142.883	0.172	0.118	44.680	124.896	0.174
	FEA	0.173	50.547	186.557	0.367	0.121	47.975	150.549	0.226	0.112	44.539	127.613	0.176
	Error	-23.08%	-1.09%	-9.20%	-24.17%	-26.71%	-1.19%	-5.09%	-23.66%	5.46%	0.31%	-2.13%	-1.00%
	Pred.	0.109	47.088	135.023	0.176	0.136	44.220	137.306	0.231	0.008	46.082	154.885	0.126
	FEA	0.109	46.171	133.503	0.178	0.133	44.071	139.675	0.236	0.136	48.248	161.653	0.272
	Error	-0.48%	1.99%	1.14%	-1.48%	1.88%	0.34%	-1.70%	-2.06%	-94.05%	-4.49%	-4.19%	-53.78%
No Surrogate	FEA	0.102	43.461	134.424	0.191	0.105	42.253	119.353	0.161	0.082	48.914	135.563	0.135
	FEA	0.087	46.145	131.496	0.148	0.095	44.553	127.989	0.158	0.088	48.349	134.457	0.142
	FEA	0.080	49.086	149.476	0.170	0.108	45.176	125.133	0.160	0.082	48.914	135.563	0.135